

Qualification Pack



Junior Technician - Mechanized Sewer Cleaning

QP Code: SGJ/Q4006

Version: 1.0

NSQF Level: 3

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SGJ/Q4006: Junior Technician - Mechanized Sewer Cleaning

Brief Job Description

Junior Technician- Mechanized Sewer Cleaning, is an individual, who is responsible for emptying, cleaning, transport and disposal of sludge from the sewer line to a specified site/Co-treatment plant. He /She has to operate and maintain Mechanized sewage cleaning machine, Sludge Vacuum tank with trucks, pumps, suction hoses, and other machinery/equipment to empty the manholes.

Personal Attributes

This job requires the individual to be responsible for Mechanized cleaning, transport and disposal of faecal sludge from the septic tank/sewer line to desludging site / FSTP / Co-treatment plant. He /She has to operate Sludge Vacuum tank with trucks, pumps, suction hoses, and other machinery/equipment to empty the septic tank./sewer line. Jr. Technician -Sewer Mechanized cleaning must be active and capable of safely handling the job role.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N4040: Introduce Faecal Sludge management and outline various opportunities for SSWs](#)
2. [SGJ/N4041: Maintain health and safety at faecal Sludge Management site](#)
3. [SGJ/N4042: Perform Sewer Cleaning with Mechanised Equipment](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Technician
Country	India
NSQF Level	3
Credits	11
Aligned to NCO/ISCO/ISIC Code	NCO-2015/9613.0500, Sweeper, Sewer

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Minimum Educational Qualification & Experience	5th grade pass with 5 Years of experience of relevant work includes in Waste/ waste water management, sanitation or in a related field. OR 8th grade pass (with two years of (NTC/ NAC) after 8th) with NA of experience OR 8th grade pass with 2 Years of experience of relevant work includes in Waste/ waste water management, sanitation or in a related field OR Previous relevant Qualification of NSQF Level (2.5) with 1-2 Years of experience of relevant work includes in Waste/ waste water management, sanitation or in a related field OR Previous relevant Qualification of NSQF Level (2) with 3 Years of experience of relevant work includes in Waste/ waste water management, sanitation or in a related field
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	11/08/2026
NSQC Approval Date	11/08/2023
Version	1.0
Reference code on NQR	QG-03-ES-00668-2023-V1-SCGJ
NQR Version	1

Remarks:

Total 330 Hours i.e. 11 Credits (90 hours theory + 120 hours practical + 30 hours employability skills + 90 hours On the Job Training (OJT)

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SGJ/N4040: Introduce Faecal Sludge management and outline various opportunities for SSWs

Description

This unit is about introduce Faecal Sludge management and outline various opportunities for SSWs

Scope

The scope covers the following :

- Mechanized Sewer Cleaning : Introduction and Key Schemes
- Provision and Entitlements for SSW- Role of ULB
- Improving Health, Education and other opportunities for SSWs

Elements and Performance Criteria

Mechanized sewer cleaning: Introduction and key schemes

To be competent, the user/individual on the job must be able to:

- PC1.** provide introduction to Faecal Sludge management and discuss the importance and challenges of mechanised sewer cleaning
- PC2.** explain the objectives and modalities of Government schemes for faecal sludge management like NAMASTE scheme, Swachhta Udyami Yojana (SUY) etc
- PC3.** explain benefits of NAMASTE or other related govt. scheme for Sewage and Septic Tank Sanitation Workers (SSW)
- PC4.** describe Legal provision for punitive actions against hazardous cleaning as per PEMSR Act, 2013
- PC5.** describe roles and responsibilities of the Employer, Local Authority in preventing Hazardous manual cleaning
- PC6.** explain roles of Emergency Response Sanitation Unit (ERSU) including Responsible Sanitation Authority (RSA) and Duty Supervisor
- PC7.** explain roles of Sewer Entry Professionals (SEP) and Private Sanitation Service Organisation (PSSO)
- PC8.** identify role of ERSU Authority, ULB in case of manual entry of SSW required
- PC9.** discuss the powers of the Sanitary Inspector as part of PEMSR Act, 2013
- PC10.** discuss roles and responsibilities of Employer, Local Authority in implementing NAMASTE scheme
- PC11.** outline the importance of Faecal Sludge management and illustrate the key challenges in mechanised sewer cleaning
- PC12.** outline opportunities and illustrate the process of availing the benefits through NAMASTE scheme and Swachhta Udyami Yojna
- PC13.** illustrate the roles and responsibilities of various stakeholders including Employers/PSSO, Local Authority/ULBs, etc in faecal sludge management

Provisions and Entitlements for Sewer/ Workers (SSW)- Role of ULBs

To be competent, the user/individual on the job must be able to:

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- PC14.** discuss various provisions & entitlements available for the SSWs through the Urban Local Bodies (ULBs)
- PC15.** explain the list of regular medical check-ups and appropriate vaccinations to be administered to SSW as a role of ULB
- PC16.** discuss about the relief package including life insurance policy for all SSW at ULBs
- PC17.** explain grievance redressal mechanism for SSW at ULB
- PC18.** discuss health assurance scheme under Pradhan Mantri Jan Arogya Yojana (PM-JAY), access to educational scholarship for children, access to housing through subsidized loans/ schemes
- PC19.** list out various parameters for medical examinations
- PC20.** show how to use the safety gears and PPE
- PC21.** list various parameters under health assurance scheme under PM-JAY and other welfare incentives
- PC22.** show how to avail health/life insurance under Pradhan Mantri Jan Arogya Yojana (PM-JAY) scheme along with accessing educational scholarship for children and access to housing through subsidized loans/ schemes
- PC23.** outline grievance redressal mechanism for SSW at ULB

Improving Health, Education and other opportunities for SSWs

To be competent, the user/individual on the job must be able to:

- PC24.** explain Swachhta Udyami Yojana (SUY) scheme including the types of loans with its financials, role of ULB's etc. and the Self-Employment Scheme for Rehabilitation of Manual Scavengers (SRMS)
- PC25.** discuss case studies on JLG/SHG formed by SSW for procurement of machines, business model and their achievements
- PC26.** discuss the steps involved in setting up a business for sewer cleaning
- PC27.** discuss how to avail direct loans by ULB as part of SUY Scheme, for procurement of sewer cleaning machines
- PC28.** discuss the process involved in the sanction of loan under SUY scheme
- PC29.** discuss the types of Enterprise model
- PC30.** discuss who are the potential customers
- PC31.** explain the dedicated scholarship schemes for Education of children
- PC32.** explain about personal family planning
- PC33.** list out various loans, welfare and promoting business opportunities for SSWs under the schemes
- PC34.** list out various parameters in the schemes for improving opportunities for family, health and education
- PC35.** illustrate key points from the case studies on JLG/SHG formed by SSW for procurement of machines etc
- PC36.** list out the traits of potential customers

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational culture and setup

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- KU2.** various policies and regulations.
- KU3.** organization's customer support policy
- KU4.** organization's documentation policy.
- KU5.** health and safety policy
- KU6.** septic tank emptying services
- KU7.** do's and don'ts of mechanized cleaning services of the septic tanks
- KU8.** no fire or flame should be used near the septic tanks as there may be inflammable gases inside septic tanks
- KU9.** installation of pipes
- KU10.** handling the vacuum tank
- KU11.** health and safety procedure
- KU12.** personal protection equipment including the use of related safety gears & equipment's
- KU13.** manual scavenging act 2013

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up relevant documents
- GS2.** read relevant documents to understand the purpose of the work
- GS3.** express statements or information clearly so that others can understand
- GS4.** participate in and understand main points of simple discussions
- GS5.** respond appropriately to queries
- GS6.** follow organization rule-based decision-making process
- GS7.** complete tasks efficiently and accurately within stipulated time
- GS8.** working constructively and collaboratively with others
- GS9.** understand limitations of the customers
- GS10.** coordinate with multidisciplinary stakeholders
- GS11.** recognize problems and search for solutions
- GS12.** choose best methods to complete assigned tasks
- GS13.** approach relevant authority when required

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Mechanized sewer cleaning: Introduction and key schemes</i>	37	15	-	-
PC1. provide introduction to Faecal Sludge management and discuss the importance and challenges of mechanised sewer cleaning	3	-	-	-
PC2. explain the objectives and modalities of Government schemes for faecal sludge management like NAMASTE scheme, Swachhta Udyami Yojana (SUY) etc	3	-	-	-
PC3. explain benefits of NAMASTE or other related govt. scheme for Sewage and Septic Tank Sanitation Workers (SSW)	3	-	-	-
PC4. describe Legal provision for punitive actions against hazardous cleaning as per PEMSAR Act, 2013	4	-	-	-
PC5. describe roles and responsibilities of the Employer, Local Authority in preventing Hazardous manual cleaning	4	-	-	-
PC6. explain roles of Emergency Response Sanitation Unit (ERSU) including Responsible Sanitation Authority (RSA) and Duty Supervisor	4	-	-	-
PC7. explain roles of Sewer Entry Professionals (SEP) and Private Sanitation Service Organisation (PSSO)	4	-	-	-
PC8. identify role of ERSU Authority, ULB in case of manual entry of SSW required	4	-	-	-
PC9. discuss the powers of the Sanitary Inspector as part of PEMSAR Act, 2013	4	-	-	-
PC10. discuss roles and responsibilities of Employer, Local Authority in implementing NAMASTE scheme	4	-	-	-
PC11. outline the importance of Faecal Sludge management and illustrate the key challenges in mechanised sewer cleaning	-	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. outline opportunities and illustrate the process of availing the benefits through NAMASTE scheme and Swachhta Udyami Yojna	-	5	-	-
PC13. illustrate the roles and responsibilities of various stakeholders including Employers/PSSO, Local Authority/ULBs, etc in faecal sludge management	-	5	-	-
<i>Provisions and Entitlements for Sewer/ Workers (SSW)- Role of ULBs</i>	21	25	-	-
PC14. discuss various provisions & entitlements available for the SSWs through the Urban Local Bodies (ULBs)	4	-	-	-
PC15. explain the list of regular medical check-ups and appropriate vaccinations to be administered to SSW as a role of ULB	4	-	-	-
PC16. discuss about the relief package including life insurance policy for all SSW at ULBs	4	-	-	-
PC17. explain grievance redressal mechanism for SSW at ULB	4	-	-	-
PC18. discuss health assurance scheme under Pradhan Mantri Jan Arogya Yojana (PM-JAY), access to educational scholarship for children, access to housing through subsidized loans/ schemes	5	-	-	-
PC19. list out various parameters for medical examinations	-	5	-	-
PC20. show how to use the safety gears and PPE	-	5	-	-
PC21. list various parameters under health assurance scheme under PM-JAY and other welfare incentives	-	5	-	-
PC22. show how to avail health/life insurance under Pradhan Mantri Jan Arogya Yojana (PM-JAY) scheme along with accessing educational scholarship for children and access to housing through subsidized loans/ schemes	-	5	-	-
PC23. outline grievance redressal mechanism for SSW at ULB	-	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Improving Health, Education and other opportunities for SSWs</i>	32	20	-	-
PC24. explain Swachhta Udyami Yojana (SUY) scheme including the types of loans with its financials, role of ULB's etc. and the Self-Employment Scheme for Rehabilitation of Manual Scavengers (SRMS)	4	-	-	-
PC25. discuss case studies on JLG/SHG formed by SSW for procurement of machines, business model and their achievements	4	-	-	-
PC26. discuss the steps involved in setting up a business for sewer cleaning	4	-	-	-
PC27. discuss how to avail direct loans by ULB as part of SUY Scheme, for procurement of sewer cleaning machines	4	-	-	-
PC28. discuss the process involved in the sanction of loan under SUY scheme	3	-	-	-
PC29. discuss the types of Enterprise model	3	-	-	-
PC30. discuss who are the potential customers	3	-	-	-
PC31. explain the dedicated scholarship schemes for Education of children	4	-	-	-
PC32. explain about personal family planning	3	-	-	-
PC33. list out various loans, welfare and promoting business opportunities for SSWs under the schemes	-	5	-	-
PC34. list out various parameters in the schemes for improving opportunities for family, health and education	-	5	-	-
PC35. illustrate key points from the case studies on JLG/SHG formed by SSW for procurement of machines etc	-	5	-	-
PC36. list out the traits of potential customers	-	5	-	-
NOS Total	90	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4040
NOS Name	Introduce Faecal Sludge management and outline various opportunities for SSWs
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Technician
NSQF Level	3
Credits	2
Version	1.0
Last Reviewed Date	NA
Next Review Date	11/08/2026
NSQC Clearance Date	11/08/2023

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SGJ/N4041: Maintain health and safety at faecal Sludge Management site

Description

This unit is about maintain health and safety at faecal sludge management site

Scope

The scope covers the following :

- Ensuring Health & Safety in Mechanised Sewer Cleaning

Elements and Performance Criteria

Ensuring Health & Safety in Mechanised Sewer Cleaning

To be competent, the user/individual on the job must be able to:

- PC1.** explain the importance of safety and discuss how to incorporate safety measures in mechanised sewer cleaning
- PC2.** discuss how to fully adhere to safety requirements and guidelines for sewer cleaning
- PC3.** explain how to ensure personal cleanliness and hygiene
- PC4.** discuss the potential hazards and preventive actions to be taken during mechanical cleaning
- PC5.** discuss mitigating occupational hazards like biological hazards, gas hazards, electrical hazards, fire hazards, chemical drain openers, heavy load lifting, high-pressure water, sinking and animal/rodent/snake bites etc.
- PC6.** discuss the common myths about PPE such as comfort and productivity
- PC7.** discuss the possible injuries from sharp objects, animal and/ or insect, bites, management of bleeding, bruises, and sprain
- PC8.** discuss how to respond to emergencies and identify the essential items in first aid box with their uses
- PC9.** explain first aid intervention during suffocation, asphyxiation, and nausea
- PC10.** outline the importance of adhering to safety measures in mechanised sewer cleaning
- PC11.** show how to fully adhere to safety requirements and guidelines for mechanised sewer cleaning
- PC12.** demonstrate how to keep workplace clean, ensure personal cleanliness and hygiene and safe for work
- PC13.** illustrate how institutional set up function to access Emergency Response Sanitation Unit (ERSU)
- PC14.** demonstrate how to use appropriate Personal Protective Equipment (PPE) while performing work
- PC15.** demonstrate correct techniques to move an injured person during an emergency
- PC16.** show how to provide first aid to a victim in case of injuries from sharp objects, animal and/ or insect bites, management of bleeding, bruises, and sprain
- PC17.** demonstrate how to measure temperature of a person and perform emergency measures
- PC18.** demonstrate how to dispose hazardous waste as per State/ Local Authority norms

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- PC19.** perform the steps to clean and disinfect material, tools, equipment, and other supplies before starting work and after completing the job
- PC20.** demonstrate a practical session of CPR on a Dummy

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational culture and setup
- KU2.** various policies and regulations
- KU3.** organization's customer support policy.
- KU4.** organization's documentation policy
- KU5.** health and safety policy
- KU6.** septic tank emptying services
- KU7.** dos and don'ts of Mechanized cleaning services of the septic tanks.
- KU8.** no fire or flame should be used near the septic tanks as there may be inflammable gases inside septic tanks
- KU9.** installation of pipes
- KU10.** handling the vacuum tank
- KU11.** health and safety procedure
- KU12.** personal protection equipment including the use of related safety gears & equipment's
- KU13.** manual scavenging act 2013

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up relevant documents
- GS2.** read relevant documents to understand the purpose of the work
- GS3.** express statements or information clearly so that others can understand
- GS4.** participate in and understand main points of simple discussions
- GS5.** respond appropriately to queries
- GS6.** follow organization rule-based decision-making process
- GS7.** take decisions with systematic course of actions and/or response
- GS8.** complete tasks efficiently and accurately within stipulated time
- GS9.** working constructively and collaboratively with others
- GS10.** recognize problems and search for solutions
- GS11.** choose best methods to complete assigned tasks

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Ensuring Health & Safety in Mechanised Sewer Cleaning</i>	45	55	-	-
PC1. explain the importance of safety and discuss how to incorporate safety measures in mechanised sewer cleaning	5	-	-	-
PC2. discuss how to fully adhere to safety requirements and guidelines for sewer cleaning	5	-	-	-
PC3. explain how to ensure personal cleanliness and hygiene	5	-	-	-
PC4. discuss the potential hazards and preventive actions to be taken during mechanical cleaning	5	-	-	-
PC5. discuss mitigating occupational hazards like biological hazards, gas hazards, electrical hazards, fire hazards, chemical drain openers, heavy load lifting, high-pressure water, sinking and animal/rodent/snake bites etc.	5	-	-	-
PC6. discuss the common myths about PPE such as comfort and productivity	5	-	-	-
PC7. discuss the possible injuries from sharp objects, animal and/ or insect, bites, management of bleeding, bruises, and sprain	5	-	-	-
PC8. discuss how to respond to emergencies and identify the essential items in first aid box with their uses	5	-	-	-
PC9. explain first aid intervention during suffocation, asphyxiation, and nausea	5	-	-	-
PC10. outline the importance of adhering to safety measures in mechanised sewer cleaning	-	5	-	-
PC11. show how to fully adhere to safety requirements and guidelines for mechanised sewer cleaning	-	5	-	-
PC12. demonstrate how to keep workplace clean, ensure personal cleanliness and hygiene and safe for work	-	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. illustrate how institutional set up function to access Emergency Response Sanitation Unit (ERSU)	-	5	-	-
PC14. demonstrate how to use appropriate Personal Protective Equipment (PPE) while performing work	-	5	-	-
PC15. demonstrate correct techniques to move an injured person during an emergency	-	5	-	-
PC16. show how to provide first aid to a victim in case of injuries from sharp objects, animal and/ or insect bites, management of bleeding, bruises, and sprain	-	5	-	-
PC17. demonstrate how to measure temperature of a person and perform emergency measures	-	5	-	-
PC18. demonstrate how to dispose hazardous waste as per State/ Local Authority norms	-	5	-	-
PC19. perform the steps to clean and disinfect material, tools, equipment, and other supplies before starting work and after completing the job	-	5	-	-
PC20. demonstrate a practical session of CPR on a Dummy	-	5	-	-
NOS Total	45	55	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4041
NOS Name	Maintain health and safety at faecal Sludge Management site
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Technician
NSQF Level	3
Credits	1
Version	1.0
Last Reviewed Date	NA
Next Review Date	11/08/2026
NSQC Clearance Date	11/08/2023

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SGJ/N4042: Perform Sewer Cleaning with Mechanised Equipment

Description

This unit is about describe process of Sewer Cleaning with Mechanised Equipment

Scope

The scope covers the following :

- Sanitation Service Request Acceptance Checklist
- Perform key steps for Mechanized Cleaning of Sewers
- Post Mechanized Cleaning of Sewers Checklist

Elements and Performance Criteria

Sanitation Service Request Acceptance Checklist

To be competent, the user/individual on the job must be able to:

- PC1.** discuss about the helpline number 14420 for enabling citizens to register any complaints regarding unsafe practices being adopted for cleaning and maintenance of sewers and septic tanks
- PC2.** determine whether PPE, Machines and safety devices are available at site
- PC3.** assess the availability and condition of the PPE, Machines and Tools. Inform to Supervisor if it is not available
- PC4.** explain the complete hazard involved in the cleaning activity and consent requirement to start cleaning activities
- PC5.** explain the checklist for accepting sanitation service request
- PC6.** explain how to find if the request is documented from where it has been received (helpline/ informal/ PSSO)
- PC7.** inform about availability of helpline number 14420 for registering any complaint by citizens
- PC8.** assess the type of cleaning required (Mechanical cleaning/Manual Cleaning) and severity of chocking
- PC9.** explain what steps that are required to be followed, if there is a requirement of Manual cleaning and workers has to get down into the sewer, in case of Mechanical cleaning
- PC10.** role of Worker and the Authority for permitting manual cleaning of sewer where the worker has to get down into the sewer
- PC11.** explain broad steps involved in manual cleaning and mechanical cleaning
- PC12.** explain the SOPs issued by MoHUA in this regard
- PC13.** explain the PPE kit necessary for undertaking manual cleaning
- PC14.** list of safety gears (O2 gas cylinder, full wader suit, etc) required for manual cleaning as per SOP
- PC15.** explain the complete hazard involved in the cleaning activity and consent requirement to start cleaning activities
- PC16.** list out the check points before cleaning of Sewer

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PC17. demonstrate different types methods for inspection of site (direct and indirect)

PC18. picture Exercise relating to steps involved in Mechanized Cleaning of Sewers

PC19. picture exercise relating to site inspection and the devices used

Perform key steps for Mechanized Cleaning of Sewers

To be competent, the user/individual on the job must be able to:

PC20. explain how to barricade the identified site and to isolate the area

PC21. explain how to stop water overflow before any activity

PC22. identify the site inspection and examination of sewers

PC23. explain the indirect method for inspection and examination of Sewers like Light & Mirror method, Sonar system, Close Conduit Camera Method etc.

PC24. identify the emergency first aid kits

PC25. discuss an overview about waste disposal site

PC26. explain the use of PPE, opening of machine holes and ventilation

PC27. explain the use of safety devices- example, multigas detector to gauge gas levels, axial blowers if the gas levels are high for ventilation

PC28. discuss the different types of Machines and their SOP for sewer cleaning bucket machine, rodding machine, hydraulically propelled devices, jetting machine, clamp shell etc. and their usage

PC29. discuss the hose pipes connecting to machines for removal of sludge (checking of hose pipes for cracks/ holes, checking the sludge levels)

PC30. show how to perform all the key steps for Mechanized Cleaning of Sewers

PC31. demonstrate how to use PPE and safety devices

PC32. show how to open the machine holes and the use of axial blowers if the gas levels are high for ventilation

PC33. demonstrate the stopping of nearby machine holes using inflatable plugs

PC34. demonstrate use of sewer cleaning bucket machine, rodding machine, hydraulically propelled devices, jetting machine, clamp shell

PC35. show how to connect hose pipe to machines for removal of sludge

Post Mechanized Cleaning of Sewers Checklist

To be competent, the user/individual on the job must be able to:

PC36. discuss how to create a checklist for Post-Mechanized Sewer Assessment and follow that for performing various activities

PC37. explain about the cleaning of PPE, Tools, and Machinery after the completion of mechanized cleaning of sewer

PC38. explain about the cleaning of any Spillage

PC39. discuss the removal of barricade post completion of cleaning activity

PC40. explain about the safe disposal at designated disposal site

PC41. explain process of cleaning Desludging tank after work

PC42. discuss various Do's and Don'ts while disposing the sludge at a designated place

PC43. show how to create a checklist for Post-Mechanized Sewer Assessment and perform all activities

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- PC44.** show how to safely Clean PPE, Tools, Machinery as well as any Spillage post the completion of mechanized cleaning of sewers
- PC45.** show how to safely remove barricade from the site after completion of cleaning activity
- PC46.** show how to safely dispose the sludge at designated disposal site
- PC47.** show how to clean Desludging tank after work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational culture and setup
- KU2.** various policies and regulations
- KU3.** organization's customer support policy.
- KU4.** organization's documentation policy
- KU5.** health and safety policy
- KU6.** septic tank emptying services
- KU7.** do's and don'ts of desludging services of the septic tanks
- KU8.** no fire or flame should be used near the septic tanks as there may be inflammable gases inside septic tanks
- KU9.** installation of pipes
- KU10.** handling the vacuum tank
- KU11.** health and safety procedure
- KU12.** manual scavenging act 2013
- KU13.** personal protection equipment including the use of related safety gears & equipment's

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up relevant documents
- GS2.** read relevant documents to understand the purpose of the work
- GS3.** express statements or information clearly so that others can understand
- GS4.** participate in and understand main points of simple discussions
- GS5.** respond appropriately to queries
- GS6.** follow organization rule-based decision-making process
- GS7.** take decisions with systematic course of actions and/or response
- GS8.** complete tasks efficiently and accurately within stipulated time
- GS9.** working constructively and collaboratively with others
- GS10.** understand limitations of the customers
- GS11.** choose best methods to complete assigned tasks
- GS12.** approach relevant authority when required
- GS13.** ask questions for better understanding.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sanitation Service Request Acceptance Checklist</i>	43	12	-	-
PC1. discuss about the helpline number 14420 for enabling citizens to register any complaints regarding unsafe practices being adopted for cleaning and maintenance of sewers and septic tanks	3	-	-	-
PC2. determine whether PPE, Machines and safety devices are available at site	3	-	-	-
PC3. assess the availability and condition of the PPE, Machines and Tools. Inform to Supervisor if it is not available	3	-	-	-
PC4. explain the complete hazard involved in the cleaning activity and consent requirement to start cleaning activities	2	-	-	-
PC5. explain the checklist for accepting sanitation service request	2	-	-	-
PC6. explain how to find if the request is documented from where it has been received (helpline/ informal/ PSSO)	3	-	-	-
PC7. inform about availability of helpline number 14420 for registering any complaint by citizens	3	-	-	-
PC8. asses the type of cleaning required (Mechanical cleaning/Manual Cleaning) and severity of chocking	3	-	-	-
PC9. explain what steps that are required to be followed, if there is a requirement of Manual cleaning and workers has to get down into the sewer, in case of Mechanical cleaning	3	-	-	-
PC10. role of Worker and the Authority for permitting manual cleaning of sewer where the worker has to get down into the sewer	3	-	-	-
PC11. explain broad steps involved in manual cleaning and mechanical cleaning	3	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. explain the SOPs issued by MoHUA in this regard	3	-	-	-
PC13. explain the PPE kit necessary for undertaking manual cleaning	3	-	-	-
PC14. list of safety gears (O2 gas cylinder, full wader suit, etc) required for manual cleaning as per SOP	3	-	-	-
PC15. explain the complete hazard involved in the cleaning activity and consent requirement to start cleaning activities	3	-	-	-
PC16. list out the check points before cleaning of Sewer	-	3	-	-
PC17. demonstrate different types methods for inspection of site (direct and indirect)	-	3	-	-
PC18. picture Exercise relating to steps involved in Mechanized Cleaning of Sewers	-	3	-	-
PC19. picture exercise relating to site inspection and the devices used	-	3	-	-
<i>Perform key steps for Mechanized Cleaning of Sewers</i>	30	24	-	-
PC20. explain how to barricade the identified site and to isolate the area	3	-	-	-
PC21. explain how to stop water overflow before any activity	3	-	-	-
PC22. identify the site inspection and examination of sewers	3	-	-	-
PC23. explain the indirect method for inspection and examination of Sewers like Light & Mirror method, Sonar system, Close Conduit Camera Method etc.	3	-	-	-
PC24. identify the emergency first aid kits	3	-	-	-
PC25. discuss an overview about waste disposal site	3	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC26. explain the use of PPE, opening of machine holes and ventilation	3	-	-	-
PC27. explain the use of safety devices- example, multigas detector to gauge gas levels, axial blowers if the gas levels are high for ventilation	3	-	-	-
PC28. discuss the different types of Machines and their SOP for sewer cleaning bucket machine, rodding machine, hydraulically propelled devices, jetting machine, clamp shell etc. and their usage	3	-	-	-
PC29. discuss the hose pipes connecting to machines for removal of sludge (checking of hose pipes for cracks/ holes, checking the sludge levels)	3	-	-	-
PC30. show how to perform all the key steps for Mechanized Cleaning of Sewers	-	4	-	-
PC31. demonstrate how to use PPE and safety devices	-	4	-	-
PC32. show how to open the machine holes and the use of axial blowers if the gas levels are high for ventilation	-	4	-	-
PC33. demonstrate the stopping of nearby machine holes using inflatable plugs	-	4	-	-
PC34. demonstrate use of sewer cleaning bucket machine, rodding machine, hydraulically propelled devices, jetting machine, clamp shell	-	4	-	-
PC35. show how to connect hose pipe to machines for removal of sludge	-	4	-	-
<i>Post Mechanized Cleaning of Sewers Checklist</i>	21	20	-	-
PC36. discuss how to create a checklist for Post-Mechanized Sewer Assessment and follow that for performing various activities	3	-	-	-
PC37. explain about the cleaning of PPE, Tools, and Machinery after the completion of mechanized cleaning of sewer	3	-	-	-
PC38. explain about the cleaning of any Spillage	3	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC39. discuss the removal of barricade post completion of cleaning activity	3	-	-	-
PC40. explain about the safe disposal at designated disposal site	3	-	-	-
PC41. explain process of cleaning Desludging tank after work	3	-	-	-
PC42. discuss various Do's and Don'ts while disposing the sludge at a designated place	3	-	-	-
PC43. show how to create a checklist for Post-Mechanized Sewer Assessment and perform all activities	-	4	-	-
PC44. show how to safely Clean PPE, Tools, Machinery as well as any Spillage post the completion of mechanized cleaning of sewers	-	4	-	-
PC45. show how to safely remove barricade from the site after completion of cleaning activity	-	4	-	-
PC46. show how to safely dispose the sludge at designated disposal site	-	4	-	-
PC47. show how to clean Desludging tank after work	-	4	-	-
NOS Total	94	56	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N4042
NOS Name	Perform Sewer Cleaning with Mechanised Equipment
Sector	Green Jobs
Sub-Sector	Waste Management
Occupation	Technician
NSQF Level	3
Credits	4
Version	1.0
Last Reviewed Date	NA
Next Review Date	11/08/2026
NSQC Clearance Date	11/08/2023

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N4040.Introduce Faecal Sludge management and outline various opportunities for SSWs	90	60	0	0	150	33
SGJ/N4041.Maintain health and safety at faecal Sludge Management site	45	55	0	0	100	22
SGJ/N4042.Perform Sewer Cleaning with Mechanised Equipment	94	56	0	0	150	33
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	12
Total	249	201	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.